Thyristor Ratings

Useful specifications of a thyristor related to its steady state characteristics as found in a typical “manufacturer’s data sheet”

Voltage Ratings

Peak Working Forward OFF state voltage (V_{DWM}): It specifics the maximum forward (i.e, anode positive with respect to the cathode) blocking state voltage that a thyristor can withstand during working. It is useful for calculating the maximum RMS voltage of the ac network in which the thyristor can be used. A margin for 10% increase in the ac network voltage should be considered during calculation.

Peak repetitive off state forward voltage (V_{DRM}): It refers to the peak forward transient voltage that a thyristor can block repeatedly in the OFF state. This rating is specified at a maximum allowable junction temperature with gate circuit open or with a specified biasing resistance between gate and cathode. This type of repetitive transient voltage may appear across a thyristor due to "commutation" of other thyristors or diodes in a converter circuit.

Peak non-repetitive off state forward voltage (V_{DSM}): It refers to the allowable peak value of the forward transient voltage that does not repeat. This type of over voltage may be caused due to switching operation (i.e, circuit breaker opening or closing or lightning surge) in a supply network. Its value is about 130% of V_{DRM}. However, V_{DSM} is less than the forward break over voltage V_{BRF}.

Peak working reverse voltage (V_{DWM}): It is the maximum reverse voltage (i.e, anode negative with respect to cathode) that a thyristor can with stand continuously. Normally, it is equal to the peak negative value of the ac supply voltage.

Peak repetitive reverse voltage (V_{RRM}): It specifies the peak reverse transient voltage that may occur repeatedly during reverse bias condition of the thyristor at the maximum junction temperature.

Peak non-repetitive reverse voltage (V_{RSM}): It represents the peak value of the reverse transient voltage that does not repeat. Its value is about 130% of V_{RRM}. However, V_{RSM} is less than reverse break down voltage V_{BRR}.

Fig 4.7 shows different thyristor voltage ratings on a comparative scale.

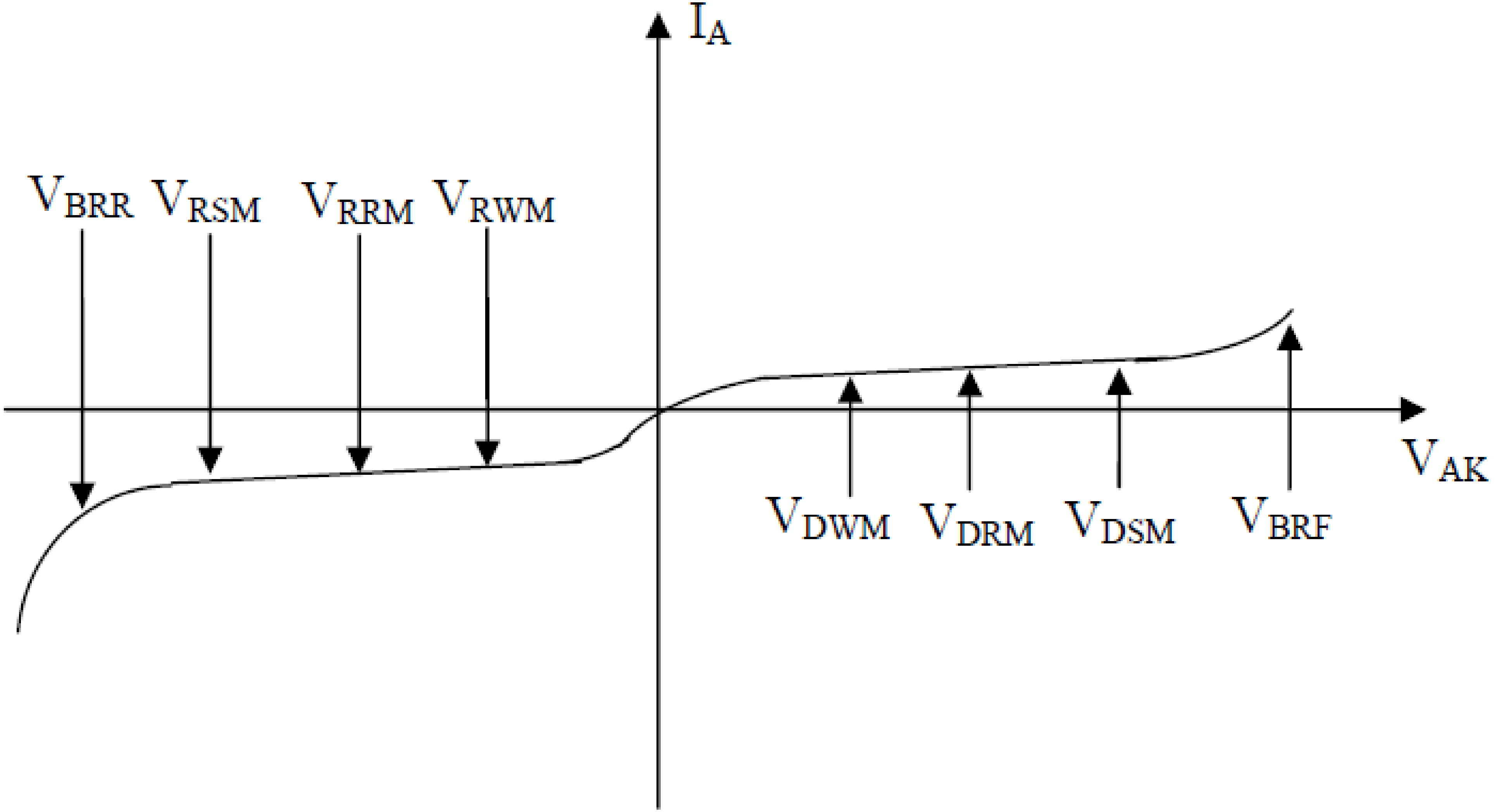

Fig. 4.7: Voltage ratings of a thyristor.

Current Ratings

Maximum RMS current (I_{rms}): Heating of the resistive elements of a thyristor such as metallic joints, leads and interfaces depends on the forward RMS current I_{rms}. RMS current rating is used as an upper limit for dc as well as pulsed current waveforms. This limit should not be exceeded on a continuous basis.

Maximum average current (I_{av}): It is the maximum allowable average value of the forward current such that

i. Peak junction temperature is not exceeded

ii. RMS current limit is not exceeded

Maximum Surge current (I_{SM}): It specifies the maximum allowable non repetitive current the device can withstand. The device is assumed to be operating under rated blocking voltage, forward current and junction temperation before the surge current occurs. Following the surge the device should be disconnected from the circuit and allowed to cool down. Surge currents are assumed to be sine waves of power frequency with a minimum duration of ½ cycles. Manufacturers provide at least three different surge current ratings for different durations.

Maximum Squared Current integral ($\int i^2 dt$): This rating in terms of A^2S is a measure of the energy the device can absorb for a short time (less than one half cycle of power frequency). This rating is used in the choice of the protective fuse connected in series with the device.